WEIGHTED DIJOIN PACKING IN ONE-EDGE DELETIONS OF CHORDAL DIGRAPHS

ARGUS OPEN RESEARCH PROJECT

Abstract. Woodall conjectured in 1976 that in every directed graph whose minimum directed cut (dicut) has size τ, there exist τ pairwise arc-disjoint dijoins. Extending the 2025 theorem of Cornuéjols, Liu, and Ravi for weighted chordal digraphs, we prove that for every finite loopless digraph $D = (V, A)$ whose underlying simple undirected graph becomes chordal after adding at most one edge, and for every nonnegative integer arc-capacity function $w\colon A \to \mathbb{Z}_{\geq 0}$ with finite minimum dicut weight $\tau(D, w)$, the maximum size $\nu(D, w)$ of a capacity-w packing of dijoins satisfies $\nu(D, w) = \tau(D, w)$.

1. Main Mathematical Results

Theorem 1.1 (Weighted dijoin packing for one-fill chordal digraphs)**.** *Let $D = (V, A)$ be a finite loopless digraph such that its underlying simple undirected graph $G(D)$ is either chordal or becomes chordal upon adding a single edge xy. Then for every nonnegative integer capacity function $w\colon A \to \mathbb{Z}_{\geq 0}$ with $\tau(D, w) < \infty$, there exist $\tau(D, w)$ dijoins of D using each arc $a \in A$ at most $w(a)$ times:*

$$\nu(D, w) \;=\; \tau(D, w).$$

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 3256 in the domain of combinatorial optimization and directed graphs adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* For every finite loopless digraph whose underlying simple graph becomes chordal upon adding at most one edge and every nonnegative integer capacity function w, the maximum w-packing of dijoins equals the minimum w-dicut weight.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted Woodall's conjecture for general digraphs

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 3256 (combinatorial optimization and directed graphs), archived in the accompanying repository package `results/3256/`.

Date: November 4, 2026.

2020 *Mathematics Subject Classification.* Primary 05C20; Secondary 90C27, 05C70.